

oh-my-paper: Deterministic Spatial PDF Fixture

oh-my-paper Research Team • Spatial Computing & Document Intelligence

Abstract

oh-my-paper is a local-first spatial canvas engineered for rigorous scientific document reading, bidirectional card linking, and offline annotation. This deterministic multi-page fixture guarantees reproducible ingestion, robust text coordinate extraction, and structured tabular validation.

1. Introduction and Architectural Motivation

Scientific inquiry demands seamless cross-referencing between hypotheses, mathematical derivations, and tabular empirical evidence. Traditional linear PDF viewers constrain cognitive synthesis. oh-my-paper treats document pages as spatially addressable artifacts on an infinite vector canvas. Users excerpt verbatim passages into atomic cards, cluster findings, and preserve provenance links.

2. Mathematical Formulations & Coordinate Transformations

Spatial projections map document space P into canvas coordinates C under scale zoom z :

Equation (1): Spatial Affine Projection

$$C(p, z, t) = z * (p - c_origin) + c_origin + [t_x , t_y]^T$$

where p is PDF coordinate, z is scale, and t is pan offset.

Equation (2): Variational Evidence Bound (ELBO)

$$L(\theta, \phi; x) = E_{\{q_{\phi}\}}[\log p_{\theta}(x | z)] - D_{KL}(q_{\phi}(z | x) || p(z))$$

optimizing retrieval likelihood under deterministic latent space regularization.

Equation (3): Multi-Head Cross Attention

$$Attention(Q, K, V) = softmax((Q * K^T) / sqrt(d_k)) * V$$

computing card relevance against document selection tokens.

Equation (4): Conservation Energy Boundary

$$\Delta E = \int_{0}^T grad f(x(t)) \cdot x_{dot}(t) dt \leq epsilon_tolerance$$

bounding energy dissipation during continuous viewport inertia animations.

3. Empirical Performance & Benchmark Results

Table 1: Execution metrics for deterministic document processing pipelines.

Component	Throughput	Latency (p95)	Memory	Status
PDF Ingestion	142 pg/sec	6.8 ms	32.4 MB	VERIFIED
Spatial Canvas	60 FPS	16.6 ms	48.1 MB	VERIFIED
Card Extraction	310 cards/s	3.2 ms	12.0 MB	VERIFIED
Coordinate Index	890 queries/s	1.1 ms	18.5 MB	VERIFIED
Cold Relaunch	N/A	24.8 ms	64.0 MB	VERIFIED

4. Concluding Verification Invariants

The evaluation confirms that deterministic fixture synthesis fulfills all structural preconditions for automated testing: fixed bounding dimensions, invariant text baseline positioning, and rigorous metadata provenance.